

AI Powered Debt Relief & Financial Recovery Platform

Phase 1

Brainstorming & Ideation

Problem Statement:

Many borrowers who default on loans or credit cards experience financial anxiety, high debt-to-income ratios, and collection pressure. They often lack financial literacy and cannot afford legal professionals to prepare hardship or negotiation letters. The project aims to build an AI-powered self-service platform that analyzes debt, predicts settlement percentages, and generates professional hardship letters using Google Gemini AI.

Empathy Map:

Target user: John Doe, a borrower facing job loss/medical hardship. The user feels stressed, hears collection calls, struggles to communicate with lenders, and seeks an affordable settlement solution.

Phase 2

Requirement Analysis

Technology Stack:

Frontend: React.js (Vite), Vanilla CSS, Lucide React.

Backend: FastAPI, Uvicorn.

Database: SQLite, SQLAlchemy ORM.

AI: Google Gemini API.

Customer Journey:

Discovery → Registration → Debt Input → Analysis → AI Drafting → Resolution.

DFD:

User submits financial details, the platform processes ratios and predictions, stores information in the database, and generates AI-powered hardship letters.

Phase 3

Project Design

Problem-Solution Fit:

- Dashboard shows DTI & EMI ratios.

- Settlement Predictor suggests negotiation amount.
- AI Generator creates hardship letters.
- Negotiation history stores previous letters.

Architecture:

React Frontend → FastAPI Backend → Financial Engine + Gemini AI → SQLite/PostgreSQL Database.

Phase 4

Project Planning

Milestones:

1. Backend setup
2. Authentication
3. Business logic
4. Gemini AI integration
5. Frontend UI
6. Testing & Deployment

Development followed sprint-based planning.

Phase 5

Project Development

Backend: FastAPI, SQLAlchemy, JWT Authentication, Financial Engine, Gemini Service.

Frontend: React Dashboard, Settlement Predictor, Letter Generator, History. Implemented features include stress meter, settlement prediction, AI letters, and audit logs.

Phase 6

Project Testing

Testing covered financial calculations, settlement prediction logic, frontend build verification, and API functionality. All automated pytest test cases passed successfully and the React application compiled without errors.

Phase 7

Project Documentation

Includes system requirements, installation guide, backend/frontend setup,

application execution, and user guide covering account creation, loan management, settlement prediction, and AI letter generation.

Phase 8

Project Demonstration

Demonstration includes login, dashboard, stress meter, settlement predictor, AI hardship letter generation, and history logs.

Future Scope:

- Bank API Integration
- Credit Score APIs
- Automated Mailing
- AI Settlement Chatbot